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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations
May 2025

CS2103-1 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT
(For CS, CB)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Explain what is meant by **PRODUCT** with reference to one of the eight principles as per the **ACM/IEEE Code of Ethics**.
A) The product should be easy to use
B) Software engineers shall ensure that their products and related modifications meet the highest professional standards possible
C) Software engineers shall ensure that their products and related modifications satisfy the client
D) It means that the product designed/created should be easily available
2. Which of the following property does not correspond to a good Software Requirements Specification (SRS) ?
A) Verifiable
B) Ambiguous
C) Complete
D) Traceable
3. Requirements should specify _____ 'but not 'how' aspects of a software.
A) 'what'
B) 'how'
C) 'why'
D) 'which'
4. Software Engineering is an engineering discipline that is concerned with?
A) how computer systems work
B) theories and methods that underlie computers and software systems
C) all aspects of software production
D) all aspects of computer-based systems development, including hardware, software and process engineering
5. The major shortcoming of waterfall model is?
A) the difficulty in accommodating changes after requirement analysis
B) the difficult in accommodating changes after feasibility analysis
C) the system testing
D) the maintenance of system
6. Project risk factor is considered in which SDLC model?
A) Incremental Model
B) Waterfall model
C) Prototyping model
D) Spiral model
7. If requirements are easily understandable and defined then which model is best suited?
A) Spiral model
B) Waterfall model
C) Incremental Model
D) Prototyping model

8. Identify which phase is not available in SDLC?
 A) Coding B) Testing
 C) Maintenance D) Abstraction
9. Which model in system modeling depicts the dynamic behavior of the system?
 A) Context Model B) Behavioral Model
 C) Data Model D) Object Model
10. _____ and _____ diagrams of UML represent Interaction modeling.
 A) Use Case, Sequence B) Class, Object
 C) Activity, State Chart D) Class, State
11. Which of the following is a complementary approach to function-oriented approach?
 A) Object oriented analysis B) Object oriented design
 C) Structured approach D) Both Object oriented analysis and design
12. Sequence of messages is emphasized by which UML diagram?
 A) state chart B) sequence
 C) activity D) collaboration
13. Agile Software Development is based on?
 A) Incremental Development B) Iterative Development
 C) Linear Development D) Both Incremental and Iterative Development
14. According to Agile manifesto what carries more value?
 A) Individuals and interactions over processes and tools B) Individuals and interactions over people and technique
 C) Individuals and interactions over projects and tools D) Individuals and interactions over products and tools
15. Verifying that whether individual software components are functioning correctly and identifying the defects in them is objective of which level of testing?
 A) Integration Testing B) Acceptance Testing
 C) Unit Testing D) System Testing
16. _____ and _____ processes are concerned with checking that software being developed meets its specification and delivers the functionality expected by the people paying for the software.
 A) Verification and Validation B) Analysis and Design
 C) Code and Design D) Analysis and Code
17. If an organization is unsure of its cost estimate, it may increase its price by a contingency over and above its normal profit. This factor is called _____.
 A) Financial Health B) Cost estimate uncertainty
 C) Market opportunity D) Requirements volatility
18. _____ is created that records the work to be done, who will do it, the development schedule, and the work products.
 A) Decision B) Solution
 C) Project plan D) Strategy
19. _____ is not a risk management activity.
 A) Risk assessment B) Risk generation
 C) Risk control D) Risk monitoring
20. Quality Management is essential for project _____.
 A) failure B) degradation
 C) delay D) success

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Why is software engineering important? Also, briefly explain with a neat diagram the various activities of requirements engineering process(RE).

Marks BT* CO* PO*

08 L*1 1 1

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|-------|---|----|----|---|---|
| b) | Explain the incremental software process model with a neat diagram and discuss its merits and demerits. | 08 | L2 | 2 | 1 |
| 2. a) | Define requirement. Also, discuss various types of non-functional requirements with suitable example for any software system of your choice. | 08 | L1 | 1 | 1 |
| b) | Explain with a neat diagram spiral software process model. | 08 | L2 | 2 | 1 |
| 3. a) | List the ACM/IEEE code of ethics that software engineers should follow to maintain professional ethical standards. Also, with example summarize an essential attributes of a good software. | 08 | L1 | 1 | 8 |
| b) | Explain the waterfall software process model with a neat diagram and discuss its merits and demerits. | 08 | L2 | 2 | 1 |

Unit – II

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|-------|--|----|----|---|---|
| 4. a) | Recall the principles of agile manifesto. | 04 | L1 | 4 | 1 |
| b) | Explain test-driven development (TDD) with a suitable diagram. | 04 | L2 | 4 | 1 |
| c) | Model the use-case diagram of a medical receptionist role in a mental health care patient monitoring system (MHC-PMS). Also, give the tabular description of transfer-data use case. | 08 | L3 | 3 | 2 |
| 5. a) | What are the advantages of pair programming? | 04 | L1 | 4 | 1 |
| b) | Explain briefly different stages in an object-oriented design process (OOD). | 06 | L2 | 3 | 1 |
| c) | Model the sequence diagram for MHC – PMS transfer data use-case. | 06 | L3 | 3 | 2 |
| 6. a) | Explain various forms of release testing with suitable example for each. | 08 | L2 | 4 | 1 |
| b) | Model the state transition diagram for microwave oven operation. Also, provide tabular representation of various states of microwave oven. | 08 | L3 | 3 | 2 |

Unit – III

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|-------|--|----|----|---|---|
| 7. a) | Show various factors affecting software pricing. | 04 | L1 | 5 | 1 |
| b) | Interpret with suitable diagram about software review process. | 04 | L2 | 5 | 1 |
| c) | Explain risk management process with suitable diagram. | 08 | L2 | 5 | 2 |
| 8. a) | Recall various advantages of a cohesive group. | 04 | L1 | 5 | 1 |
| b) | Illustrate the need of any four static software product metrics. | 04 | L2 | 5 | 1 |
| c) | Explain project scheduling process with suitable diagram. | 08 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
